

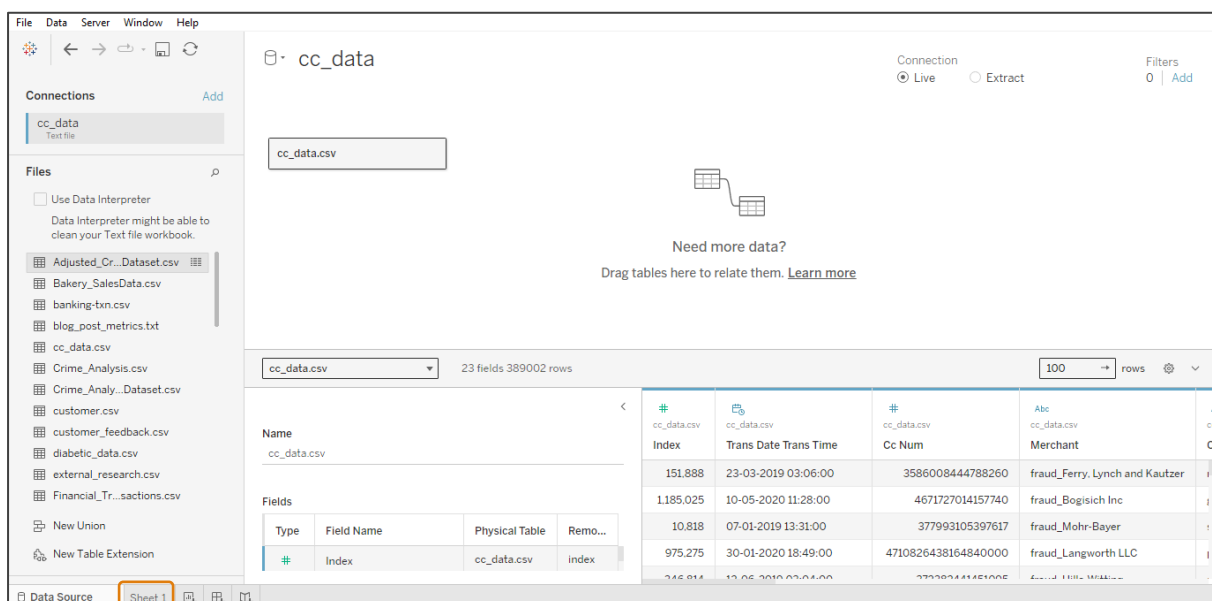
Answer Key

Task 1: Show the amount spent by different genders on different categories though a box and whisker plot

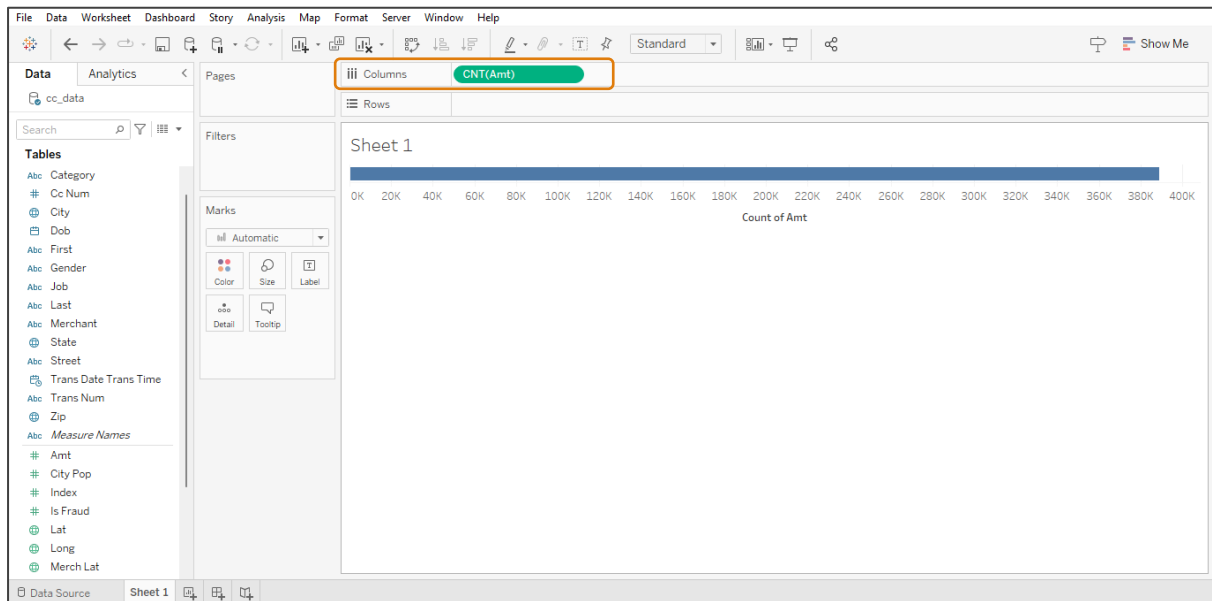
Step 1: Choose **Text file** in Connect pane and upload the **cc_data.csv** dataset



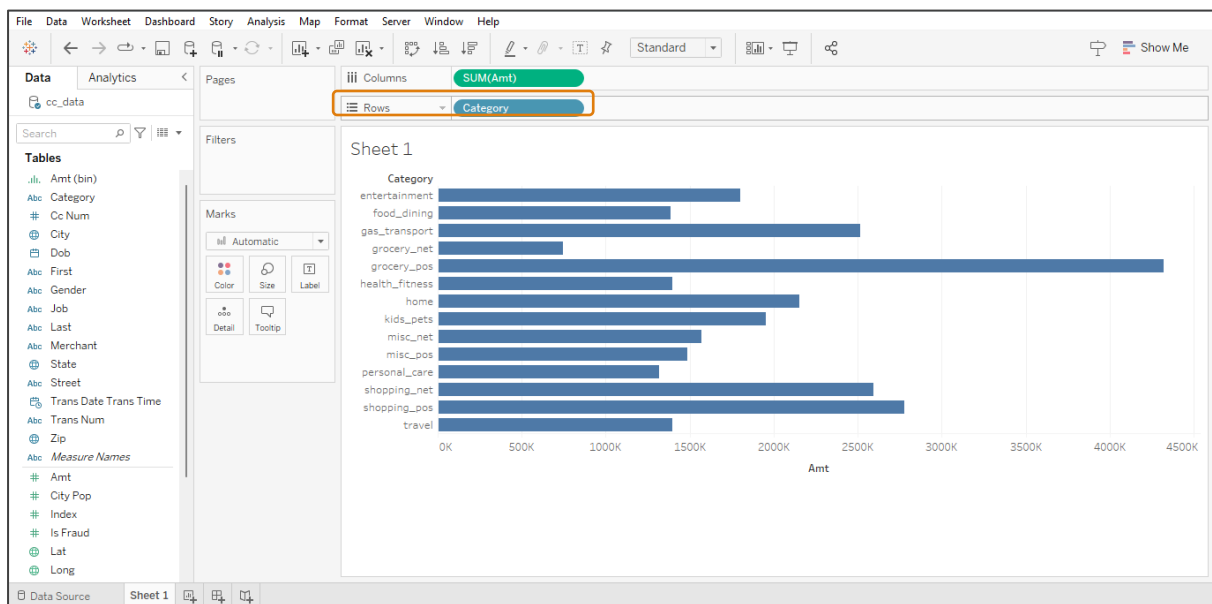
Step 2: Open a new sheet



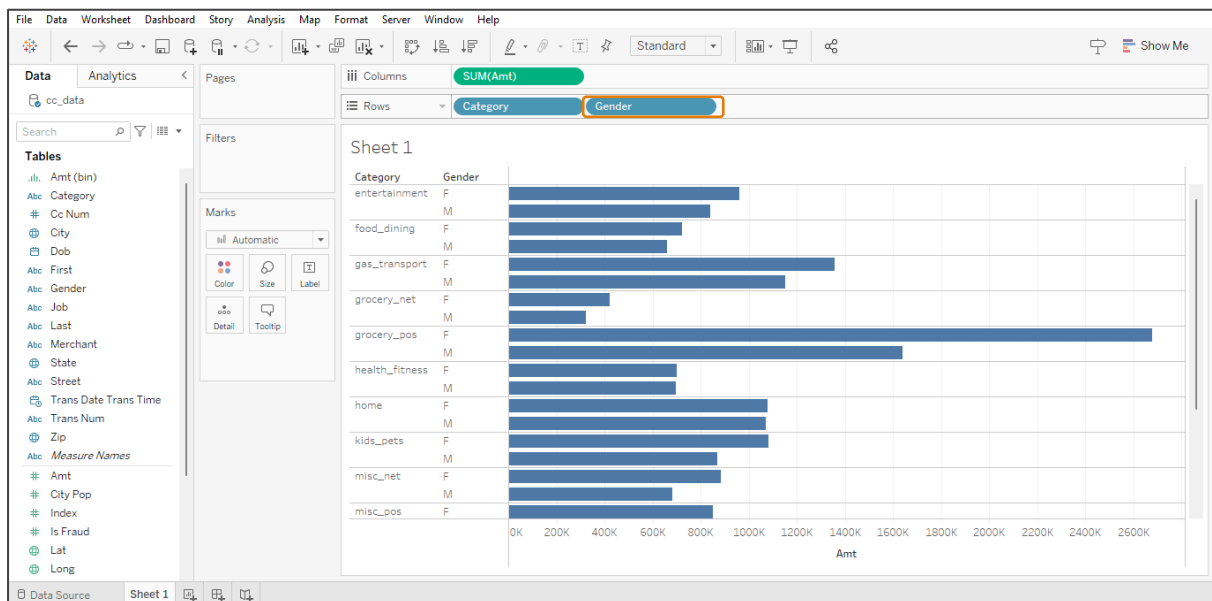
Step 3: Drag **Amt** to **Columns**



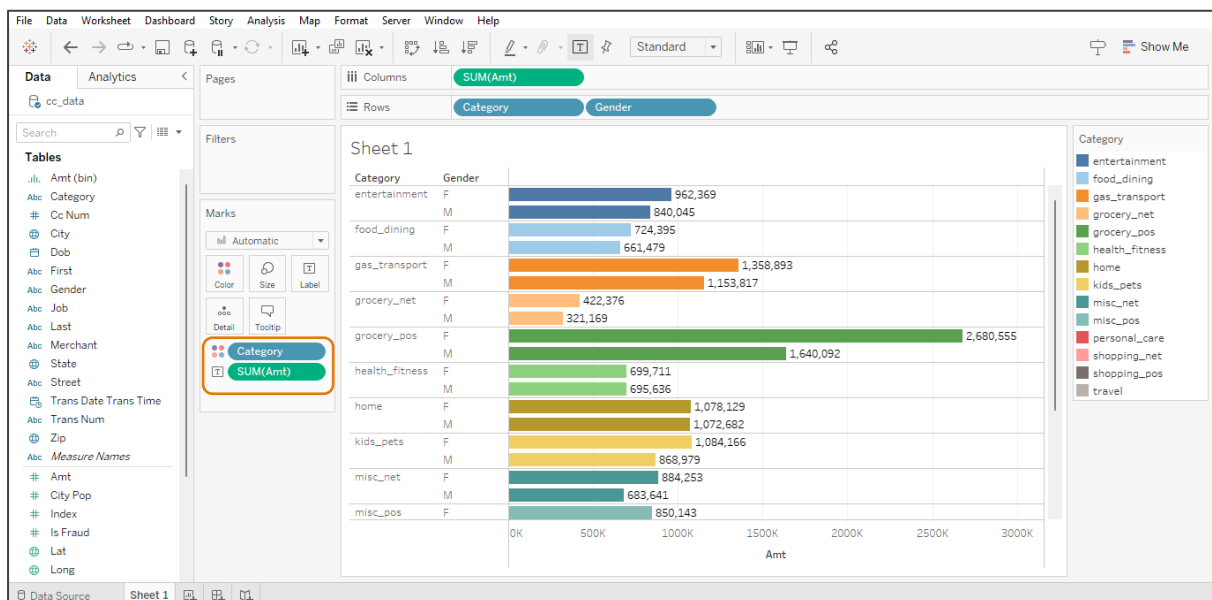
Step 4: Drag **Category** to **Rows**



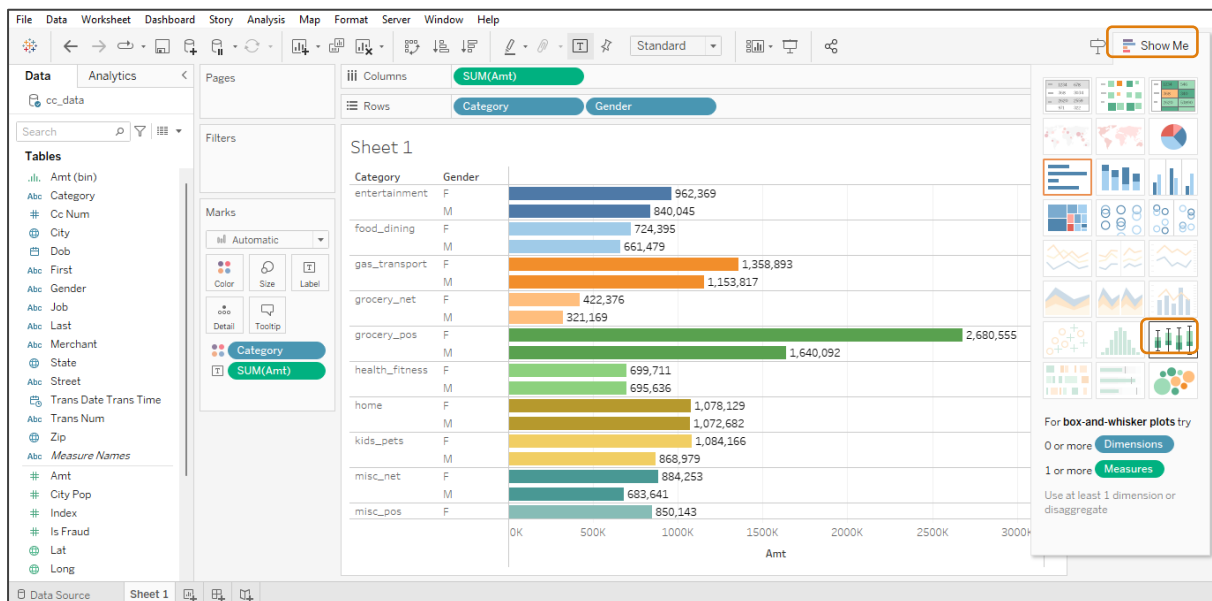
Step 5: Drag **Gender** to Rows



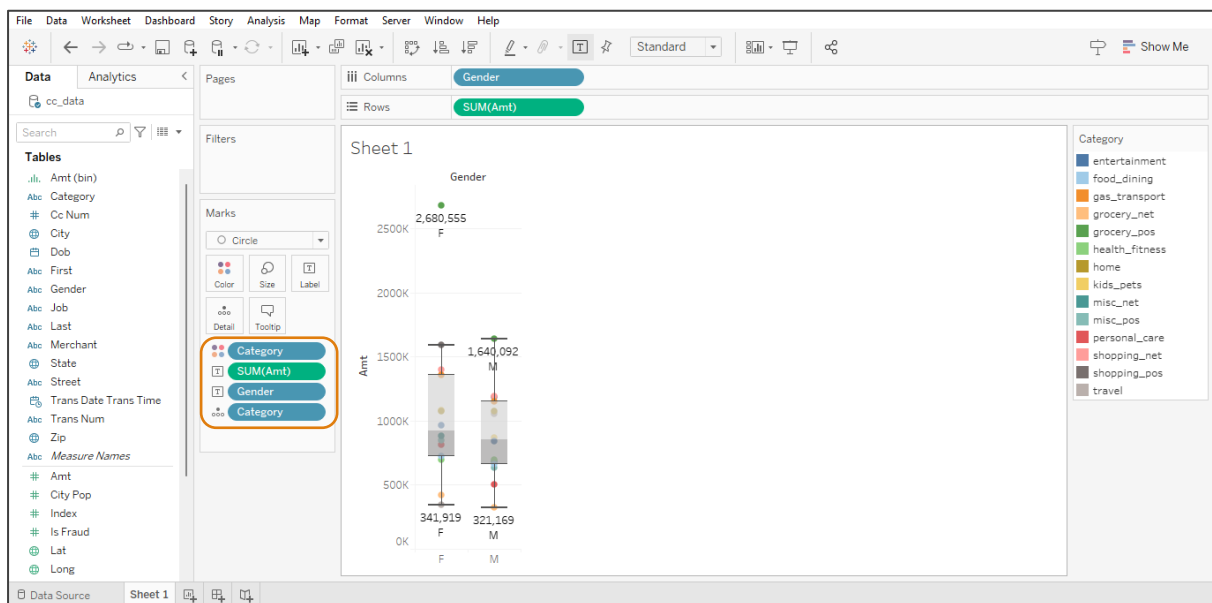
Step 6: Drag **Category** to Color and **Amt** to Label



Step 7: Choose **box** and **whisker** plot from **Show Me**



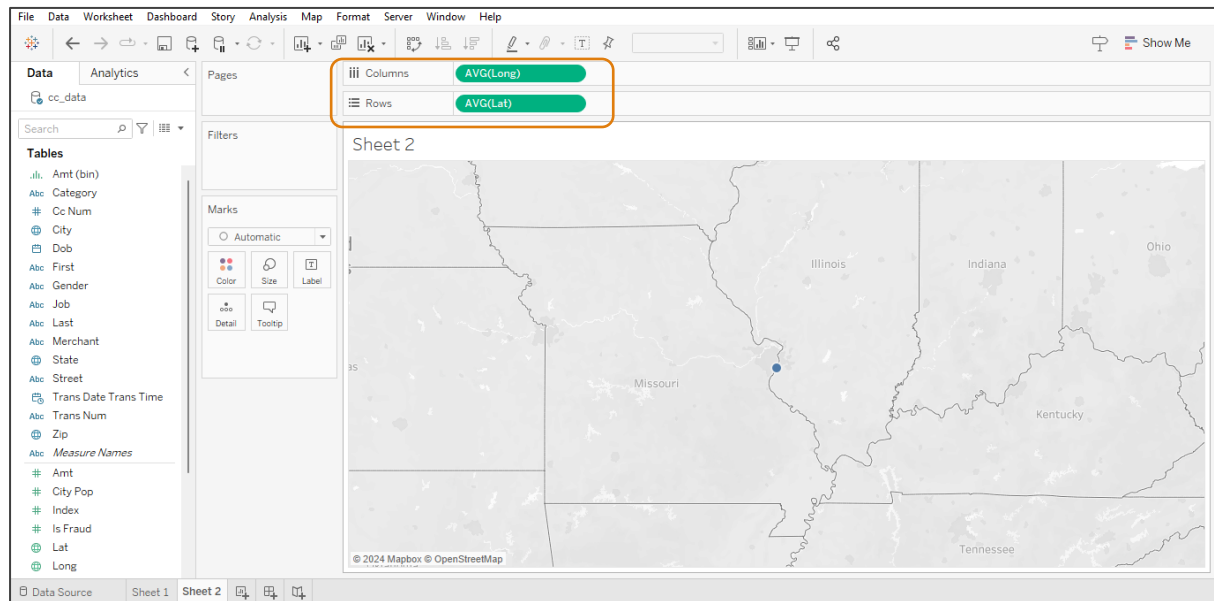
Step 8: Drag **Category** into **Color** and **Amt** into **Label**



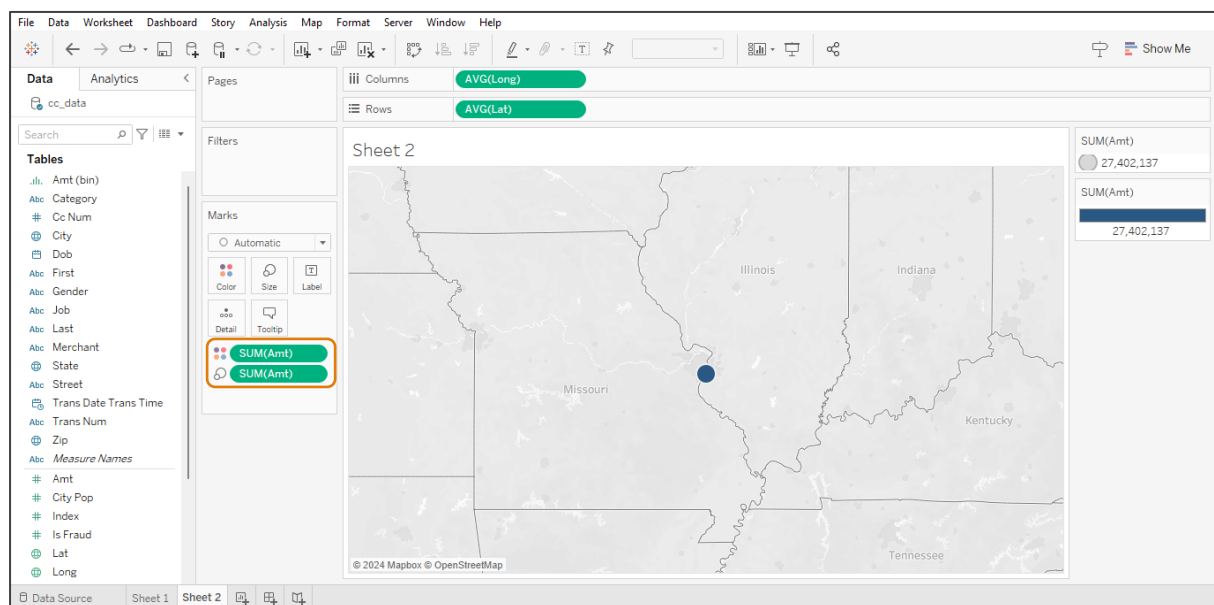
Note: Name this sheet as **Box and whisker plot**

Task 2: Show the geographical distribution of transactions using latitude and longitude coordinates

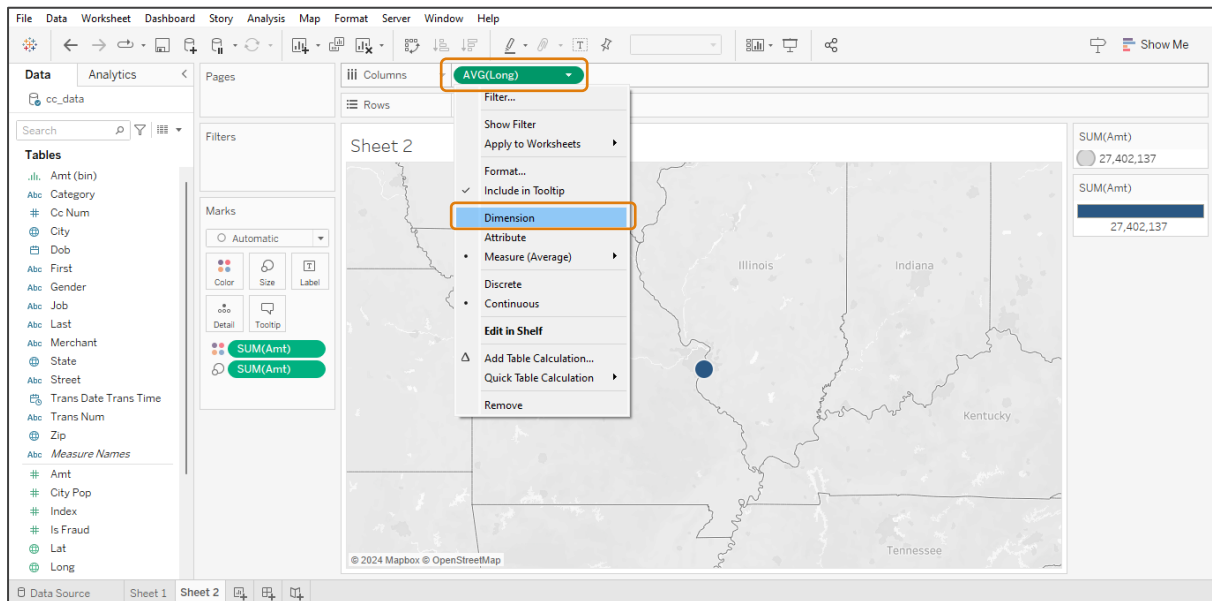
Step 1: Open a new worksheet and drag **Long** to **Columns** and **Lat** to **Rows**



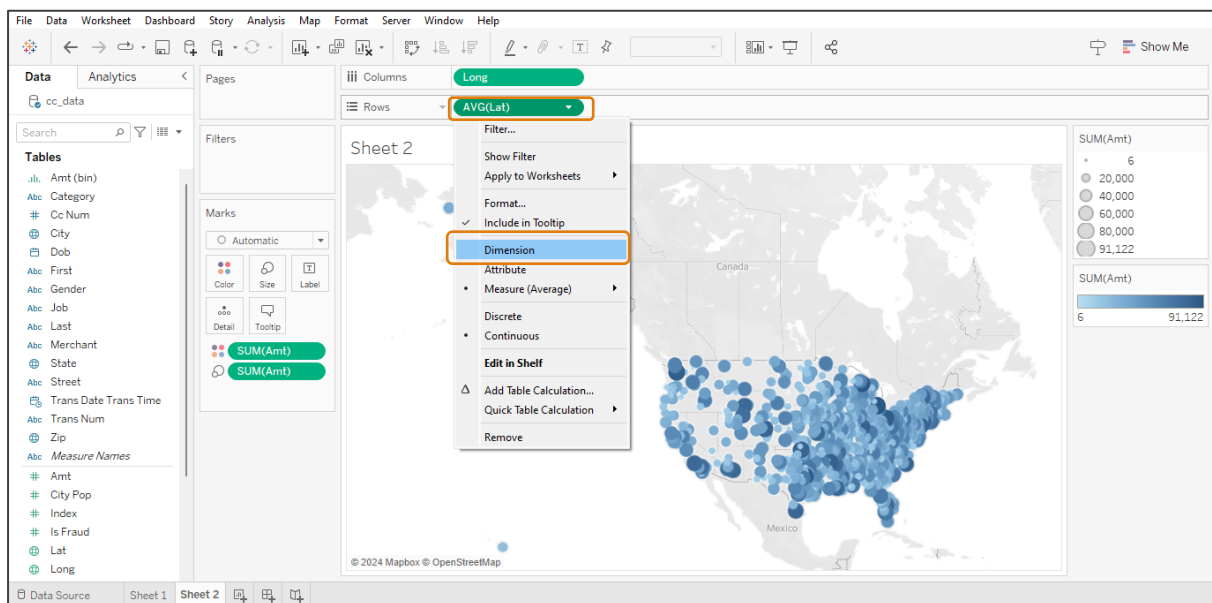
Step 2: Drag **Amt** to **Color** and **Size**



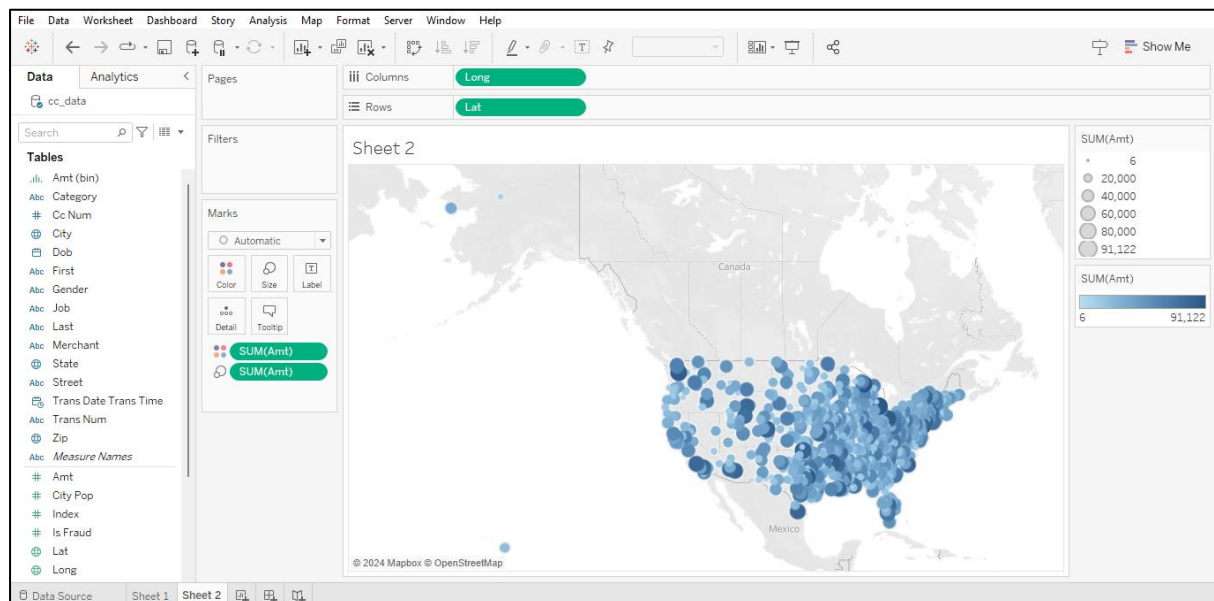
Step 3: Right click on the **Long** pill and click on **Dimension**



Step 4: Right click on the **Lat** pill and click on **Dimension**



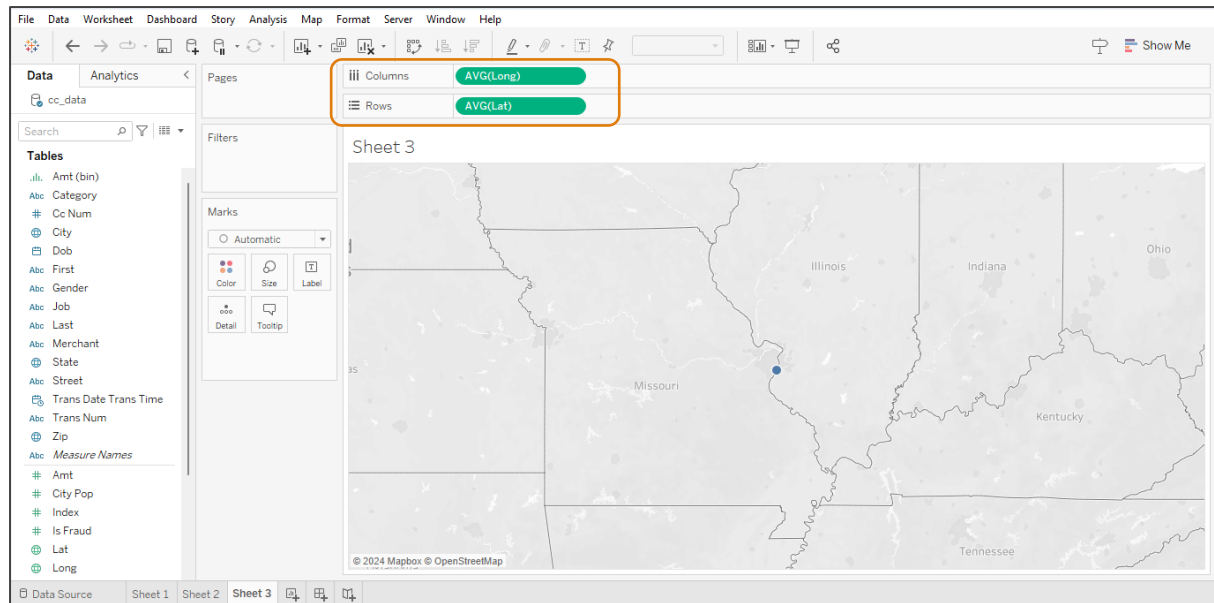
The final output will appear as below:



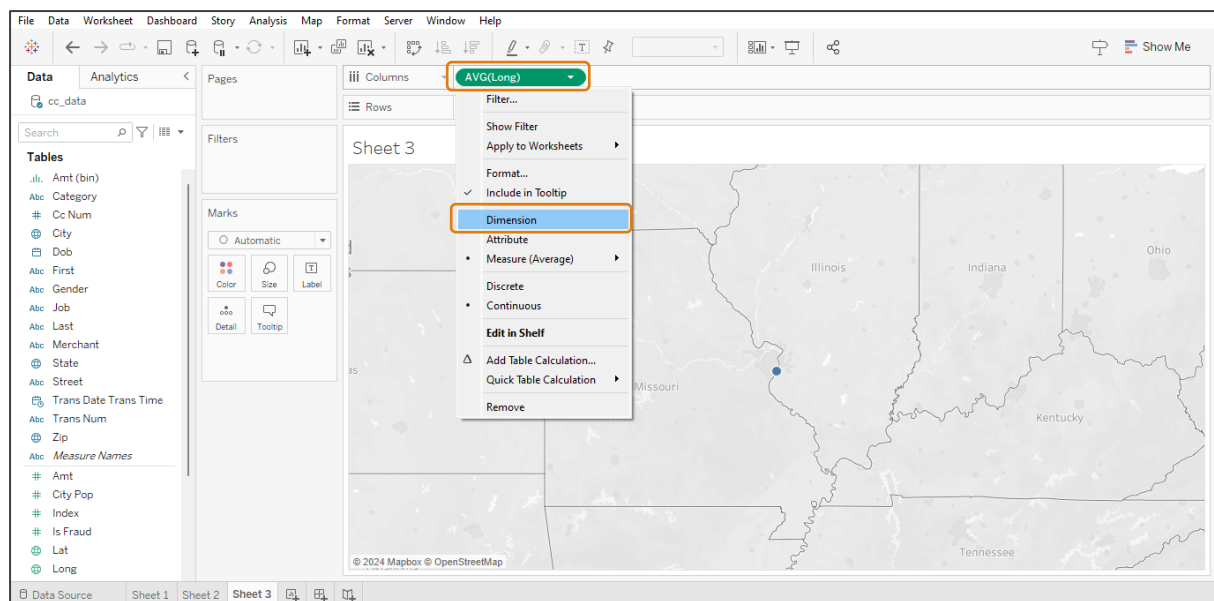
Note: Name this sheet as **Map 1**

Task 3: Use Tableau's built-in maps to visualize the locations of fraudulent transactions on a map

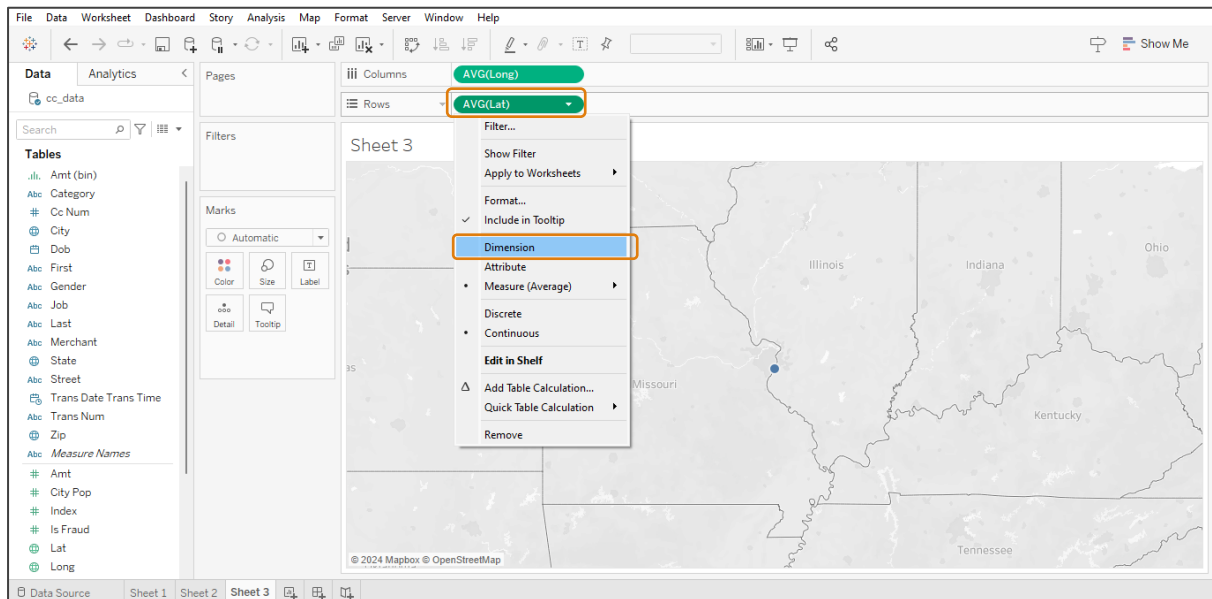
Step 1: Open a new worksheet and drag **Long** to **Columns** and **Lat** to **Rows**



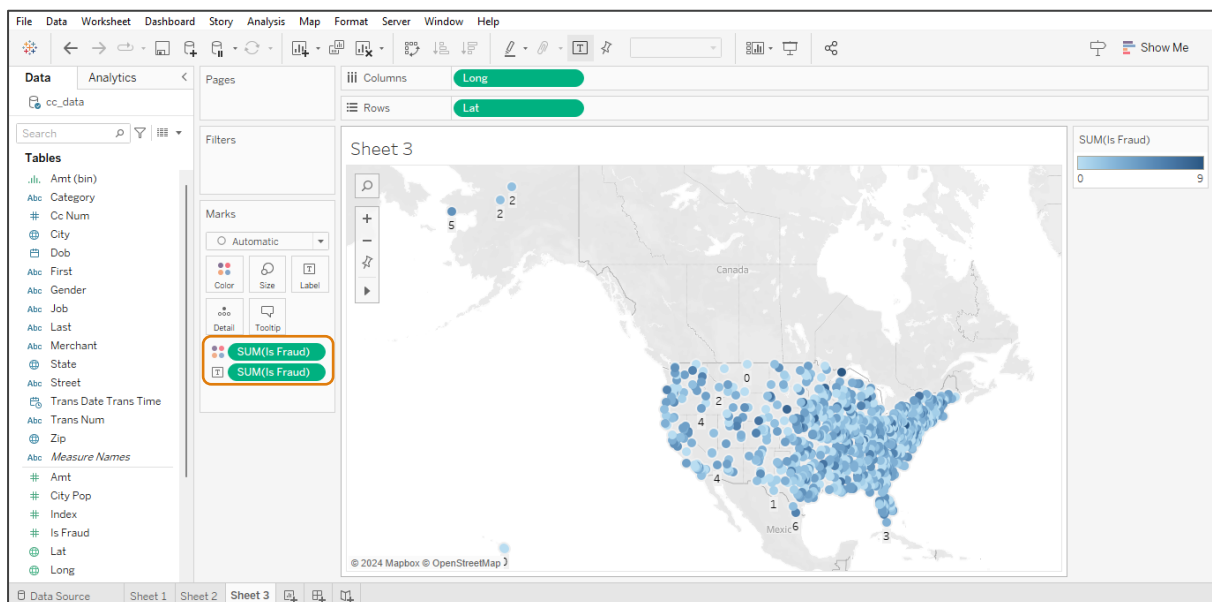
Step 2: Right click on **Long** pill and click on **Dimension**



Step 3: Right click on **Lat** pill and click on **Dimension**



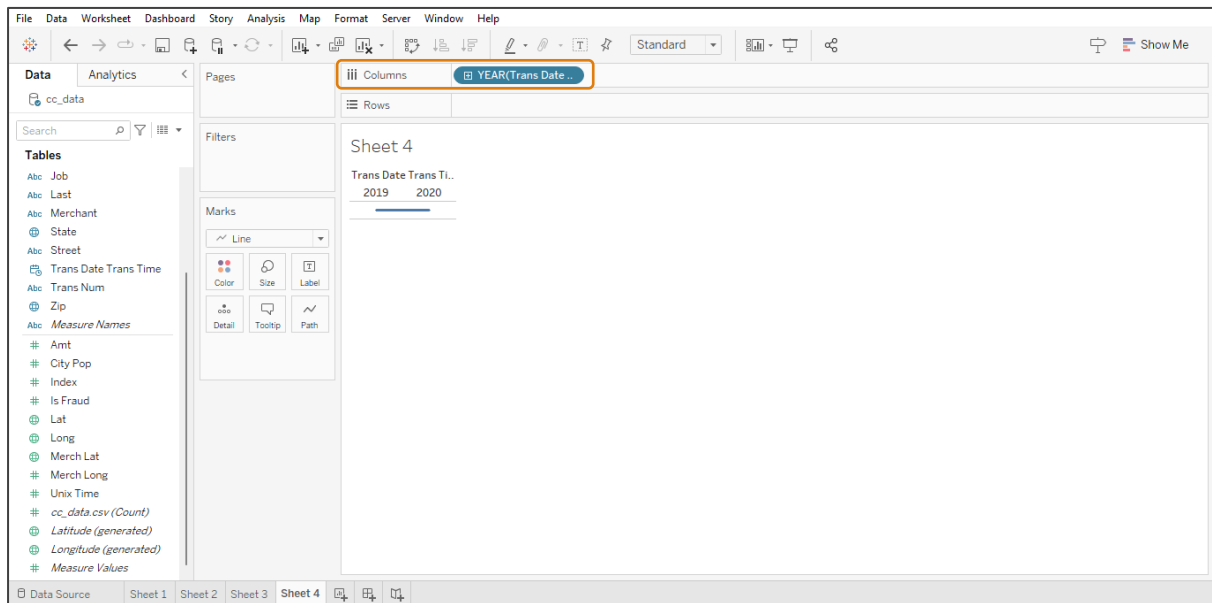
Step 4: Drag **Is Fraud** to **Color** and **Label**



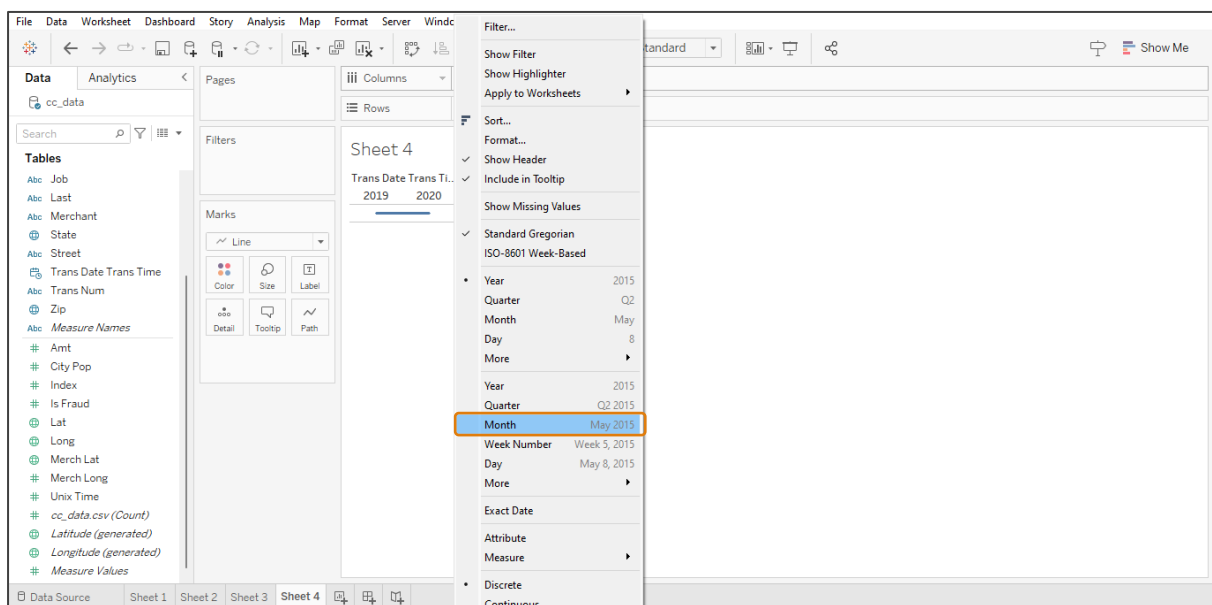
Note: Name this sheet as **Fraud Map**

Task 4: Create a time series analysis line chart showing the trend of monthly transaction amounts

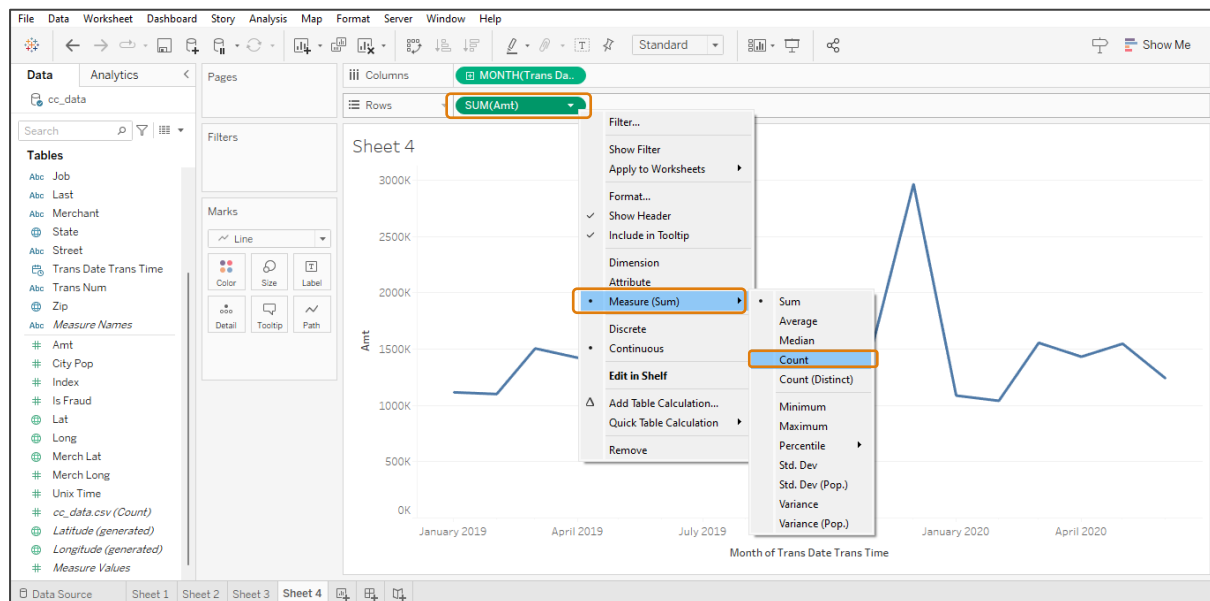
Step 1: Open a new worksheet and drag **Trans Date Trans Time** to **Columns**



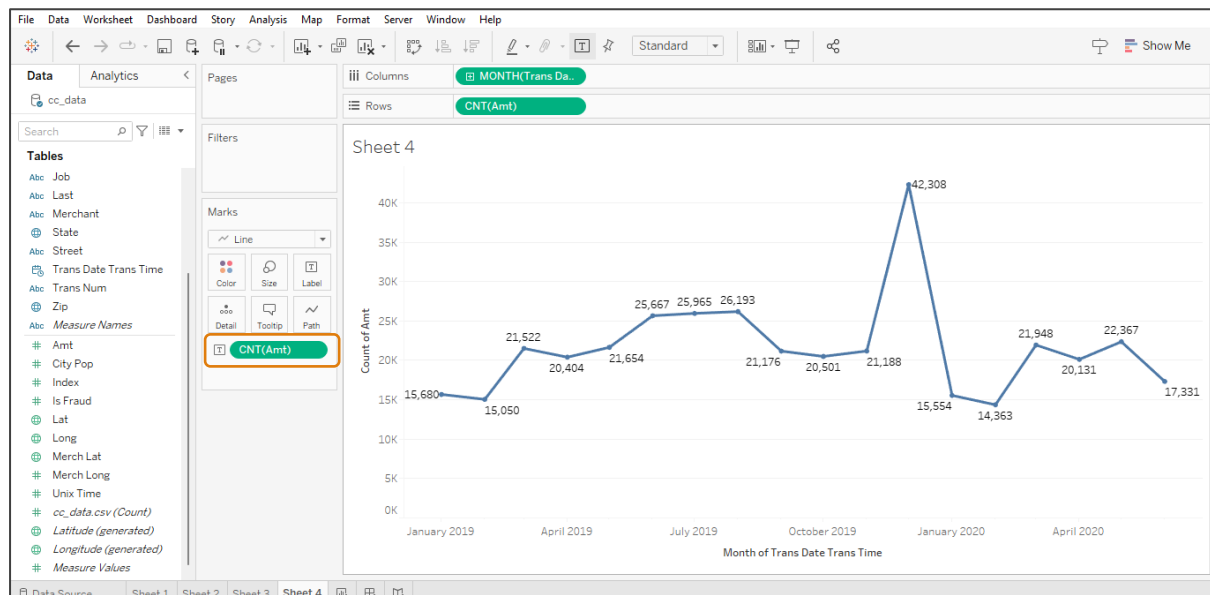
Step 2: Right-click on **Trans Date Trans Time** pill and select **Month**



Step 3: Drag **Amt** to **Rows**, right-click on it and choose the **Measure** as **Count**



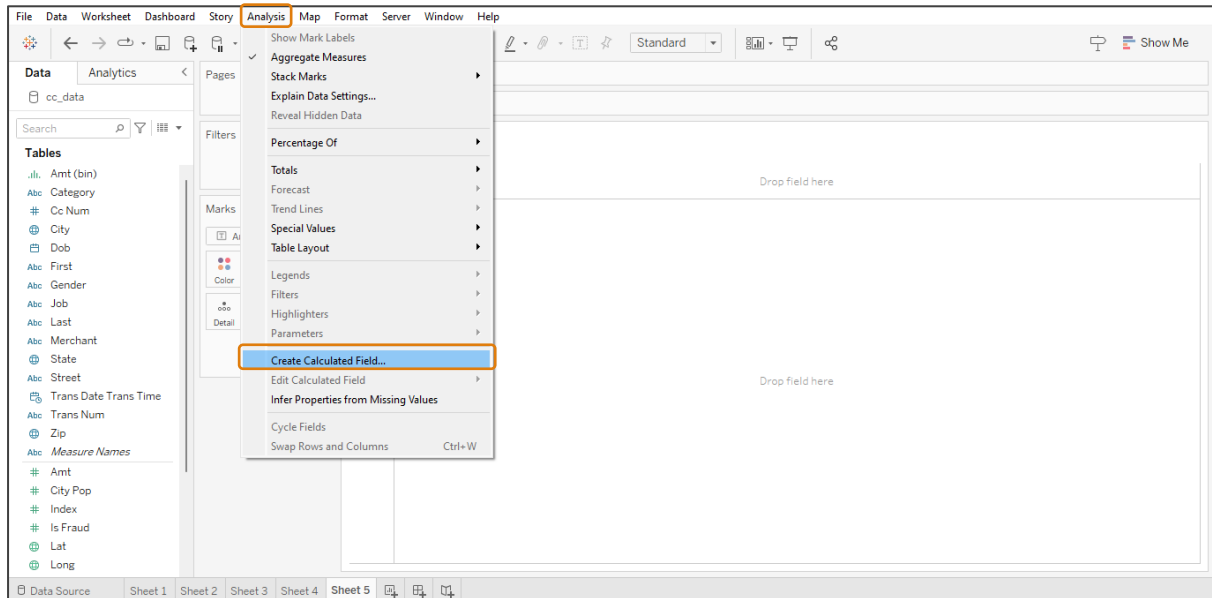
Step 4: Drag **Amt** to **Label** and change its measure to count as you have done in previous step



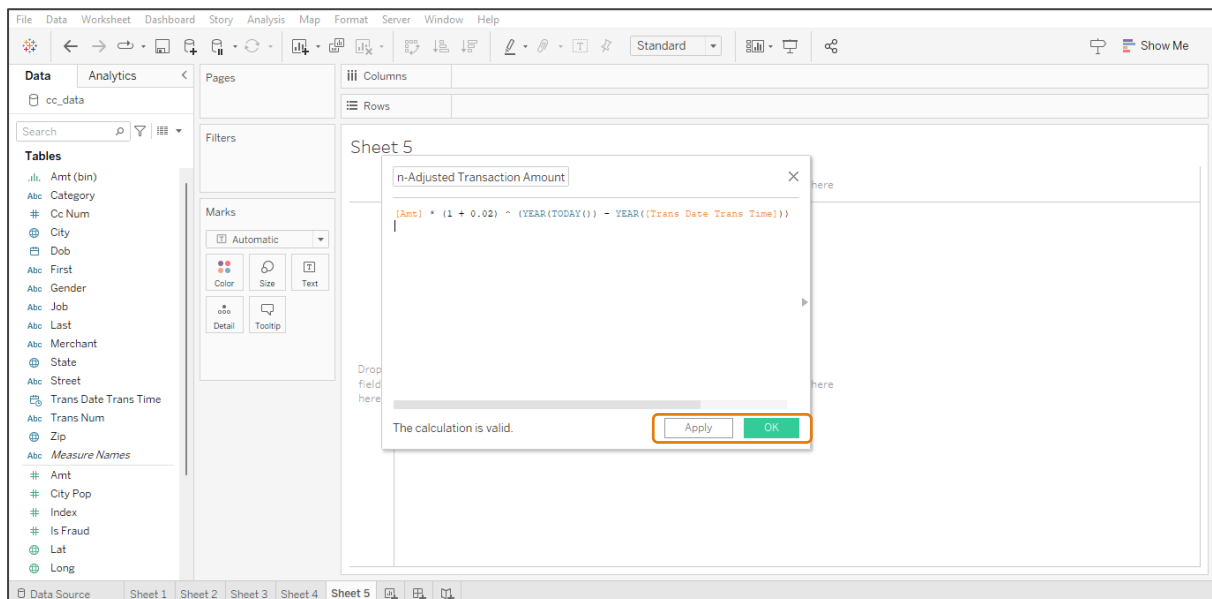
Note: Name this sheet as **Time Series**

Task 5: Create a Calculated Field for Inflation-Adjusted Transaction Amounts

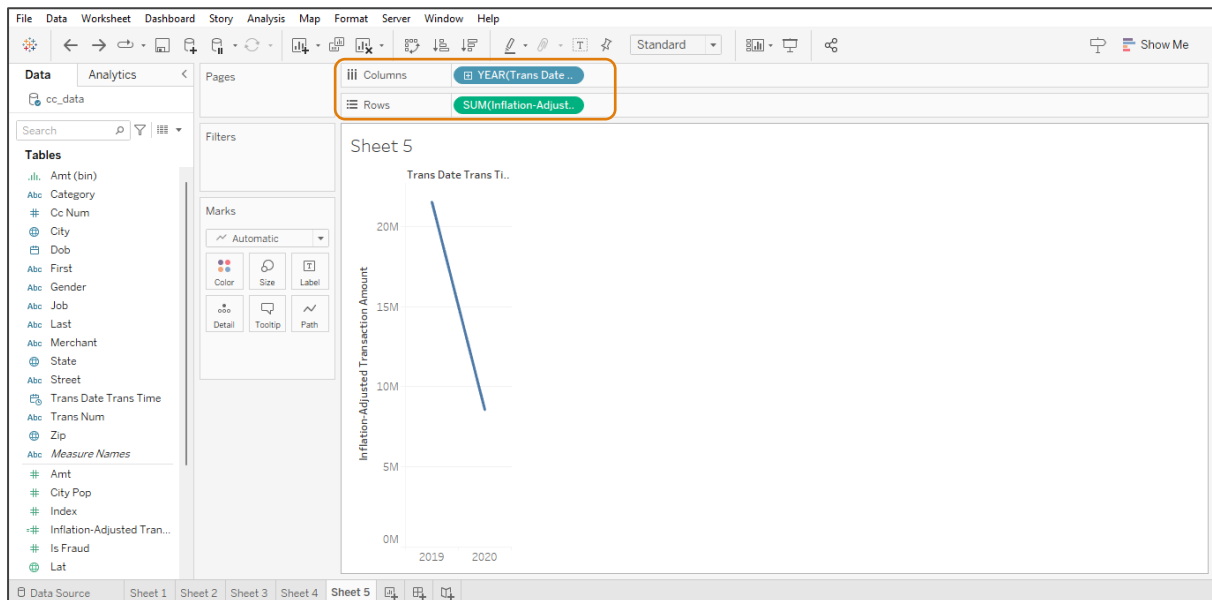
Step 1: Open a new sheet, click on **Analysis**, and choose **Create Calculated Field**



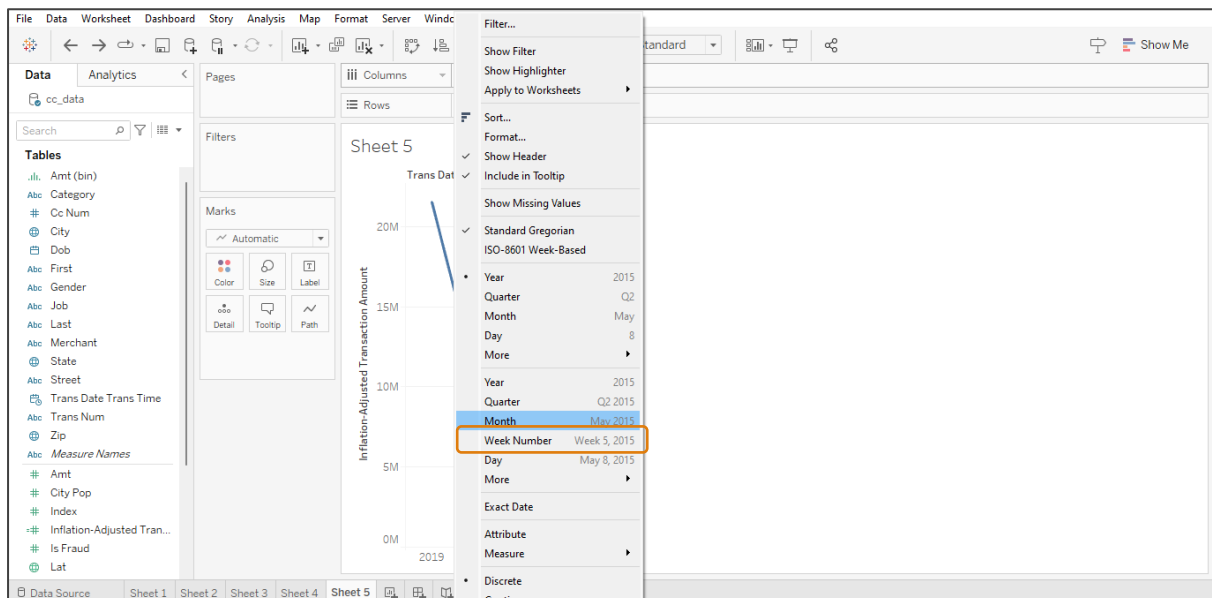
Step 2: Name the calculated field as **Inflation-Adjusted Transaction Amounts**, use the formula: **[Amt] * (1 + 0.02) ^ (YEAR(TODAY()) - YEAR([Trans Date Trans Time]))** and click on **Apply** and **OK**



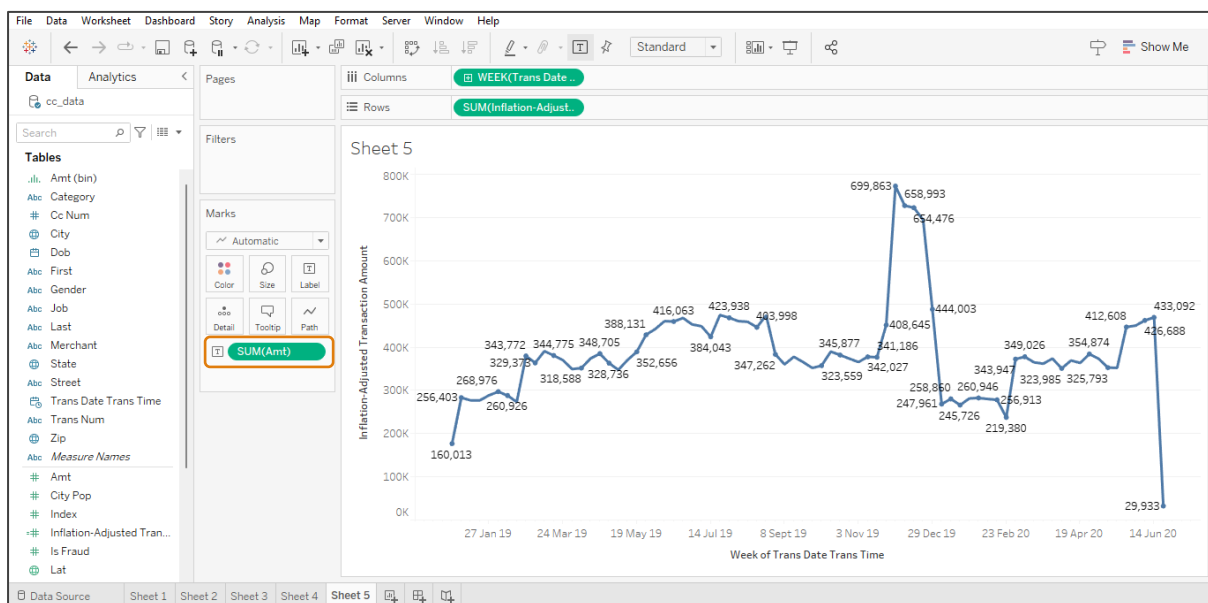
Step 3: Drag **Trans Date Trans Time** to **Columns** and **Inflation-Adjusted Transaction Amounts** to **Rows**



Step 4: Right-click on **Trans Date Trans Time** and select **Week Number**



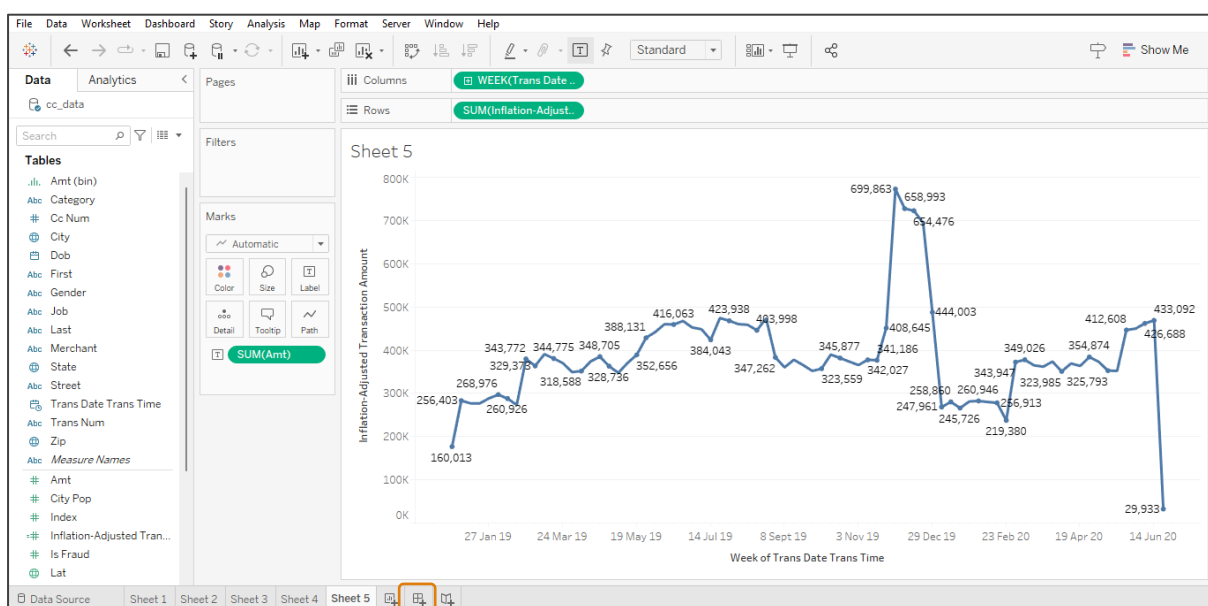
Step 5: Drag **Amt** to **Label**



Note: Name this sheet as **Inflation-Adjusted Transaction Amounts**

Task 6: Create a dashboard with all the visualizations

Step 1: Click on new dashboard icon at the bottom



Step 2: Drag **Box and whisker plot**, **Map 1**, **Fraud Map**, **Time Series** and **Inflation-Adjusted Transaction Amounts** to the dashboard area

